Rishi Bala

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Education

The University of Texas at Austin, Austin, Texas — August 2025 – May 2029

- GPA: 3.97/4, BS: Electrical and Computer Engineering Honors, BBA: Canfield Business Honors, ACT: 36/36
- **Relevant Coursework**: Computer Architecture, Digital Logic Design, Circuit Theory, Introduction to Embedded Systems

Experience

IPC Systems | New York, New York — May 2026 – August 2026

Embedded Firmware Intern

- Replaced polling-based turret screenshot delivery with live streaming by benchmarking VNC and WebRTC pipelines on embedded Linux BSP hardware, measuring end-to-end frame latency to drive a production protocol decision for financial trading floors.
- Instrumented the embedded turret diagnostics pipeline to surface real-time performance telemetry across trading floor endpoints.
- Extended the Rust/Tauri hardware interface with WAMP command surfaces, bridging the turret BSP to a desktop viewer.
- Collaborated with senior engineers to architect a hardware-software stack spanning turret-side BSP daemons (C/C++, GStreamer, Wayland), encrypted transport, and the Rust viewer application, evaluating end-to-end latency for a production decision.

Longhorn Racing – Electric | Austin, Texas — August 2025 – Present

Controls System Lead (Competing in the Formula SAE International Competition)

- Implemented real-time torque arbitration, torque vectoring, and traction control with deterministic CAN I/O and BSPD safety logic.
- Built STM32 embedded control and telemetry systems with CAN networking, multi-sensor interfacing, and real-time firmware.
- Integrated safety interlocks, fault detection, and watchdog logic across all vehicle subsystems for competition reliability.
- Designed an automotive-grade wiring harness, including connector selection, crimping, CAN routing, and full-vehicle integration.
- Coordinated cross-functional integration with mechanical and high-voltage subteams, logging 100+ miles of vehicle testing before unveiling and achieving a top-5 finish at the Formula SAE International Competition.

Projects | www.rishibala.com

FPGA-Based CME Market Making Engine — April 2026 – June 2026

Github Repository

- Designed a fully pipelined, CPU-free market maker in SystemVerilog on a Xilinx Artix-7 with a complete Ethernet/TCP/IP network stack, CME MDP 3.0 market data, and iLink 3 FIXP order entry across 25 RTL modules; sub-µs logic-pipeline tick-to-order latency.
- Implemented a CME MDP 3.0 market data pipeline decoding SBE-encoded UDP multicast packets across incremental and snapshot feeds, with a 10-level order book in RTL featuring gap detection and automatic snapshot-based sequence recovery.
- Built a quoting engine computing VWAP mid-price via a pipelined 64-bit integer divider, adjusting bid/ask quotes for inventory skew and an exponentially-decaying Order Flow Imbalance (OFI) signal derived from real-time aggressor trade data with sub.

Vehicle Control Unit (VCU) — July 2026 – Present

Github Repository | *Hardware Pictures* (fill in links later)

- Designed the Vehicle Control Unit; managed torque arbitration, BSPD logic, drivetrain control, and real-time vehicle coordination.
- Implemented torque vectoring, traction control algorithms, and thermal monitoring to improve vehicle stability and performance.
- Architected low-latency embedded firmware and deterministic CAN communication for real-time control and telemetry.

Thermal Sensor Module (TSM) — August 2025 – April 2026

Github Repository | *Hardware Pictures*

- Led the development of the TSM, a module combining radiator fan control with integrated cooling loop and flow rate monitoring.
- Owned the entire design process, including component selection, schematic design, soldering, testing, and firmware development.
- Wrote latency-optimized firmware using FreeRTOS on top of an STM32 microcontroller; developed a CAN-based telemetry system.

YerraGlasses — December 2025 – March 2026

Github Repository | *Hardware Pictures*

- Designed ESP32-S3 audio-augmented smart glasses with dual I2S, real-time ML object detection, and low-latency auditory feedback.
- Developed a space-constrained PCB with USB-C power delivery and rail regulation for a low-latency vision-to-audio pipeline.

Additional Information and Skills

Technical Skills: SystemVerilog, Verilog, FPGA, Vivado, C, C++, Rust, FreeRTOS, STM32, CAN, PCB Design, Low-Latency Systems
Honors: University Honors 2x, VEX Robotics World Finalist, World Division Champion, National Merit Commended Scholar
Interests: Pickleball, Motorsport, Drums, Jazz, Tennis, Go-Karting
Work Eligibility: Eligible to work in the U.S. with no restrictions